Factorization memory in the Clebsch hexagon code: forgetting, cubic orientation, and parent reconstruction

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Abstract

We study when the maximum-distance syndrome geometry of a projective MDS code remembers the projective system that produced it. Among six-arcs of $\mathrm{PG}(2,11)$, conic containment of the projective deep-hole locus characterizes the Clebsch hexagon and recovers its A_5 stabilizer. We place this rigidity theorem in the rank-three Coxeter family A_3, B_3, H_3: for Coxeter number h, the reflection-complement code has a common exact distance law, and at $q = h + 1$ its support is the full rational conic.

The conic child nevertheless forgets how its points were paired into secants. Differences of canonically scaled secant products are divisible by the conic equation and define quotient configurations of ranks $3, 6, 10$. For B_3 and H_3, the two factorization sheets are the unique complementary halves with equal first and second moments. Their signed moments vanish in degrees one and two, while the first surviving moment is cubic and realizes the outer orientation character. In the H_3 case, $A_4 \backslash \mathrm{PGL}_2(11)/A_5$ yields six depth profiles of sizes $1, 4, 6/1, 4, 6$; the singleton profiles recover the matching and, with its decoration, the Clebsch parent.

We finally determine which natural passages retain this orientation and which erase it. Cross-sheet designs yield certified quadratic-residue and perfect-code shadows, and golden reduction obeys an exact visible/fused law modulo 40, whereas finite theta signatures and fixed-party quantum equivalence do not distinguish the sheets. The structural statements are Lean-formalized at their stated boundaries, and the finite claims have deterministic certificates and independent replay.

Keywords. Clebsch hexagon; MDS code; deep hole; finite projective plane; arc; conic; reflection arrangement; factorization; orientation torsor.

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1 Introduction

Let C be a linear $[n, k, d]_q$ code with covering radius ρ and full-rank parity-check matrix H. For a received word v, write $\sigma(v) = Hv^{\mathsf{T}}$ and

$$w(s) = \min\{\mathrm{wt}(e) : He^{\mathsf{T}} = s\}.$$

Then $d(v, C) = w(\sigma(v))$, and v is a *deep hole* precisely when this minimum is ρ. The syndrome s specifies one coset of C, whose coset leaders are its weight-$w(s)$ representatives. Because $w(\lambda s) = w(s)$ for $\lambda \neq 0$, maximum-distance syndrome directions form a projective locus

$$\mathcal{D}_{\mathrm{proj}}(C) = \{[s] \in \mathrm{PG}(n - k - 1, q) : w(s) = \rho\}.$$

Thus one projective direction represents $q - 1$ distinct nonzero syndrome cosets, while each coset contains $|C|$ received words.

For an MDS code of redundancy $r = n - k$, the parity-check columns form a projective arc in $\mathrm{PG}(r-1, q)$, and $w(s)$ is the least number of columns whose span contains $[s]$. Adjoining a new column preserves the MDS property exactly when its direction avoids every hyperplane spanned by $r-1$ existing columns. In the redundancy-three plane case used here, this specializes to weights one, two, or three according as the direction lies on the arc, on a secant of the arc, or on neither; an extension point is precisely a direction missed by every secant. We write $\mathcal{U}(A)$ for this extension locus. If $\mathcal{U}(A) \neq \emptyset$, then the associated code has covering radius three and $\mathcal{U}(A) = \mathcal{D}_{\mathrm{proj}}(C)$. This dictionary is classical in the theory of complete arcs and is developed for MDS codes by Davydov, Marcugini, and Pambianco [10]; their Theorem 6.3 supplies the leader count used in Section 4. See also Hirschfeld [18] for the plane arc/covering-radius setting.

Henceforth *uncovered locus* and $\mathcal{U}(A)$ are the primary terms; *projective deep-hole syndrome locus* is their coding-theoretic synonym. We call a line through two points of A a *chord*; *secant* emphasizes its incidence with A, while *edge* refers to the corresponding pair in the complete graph on the arc labels and *join* emphasizes construction from its two endpoints.

The three correspondences used throughout are

$$\begin{array}{rcl} \text{parity-check column ray} & \longleftrightarrow & \text{arc point,} \\ \text{non-arc point } x \text{ on } i \text{ secants} & \longleftrightarrow & i \text{ weight-two leaders per syndrome above } x, \\ \text{uncovered point} & \longleftrightarrow & \text{projective deep-hole direction.} \end{array}$$

The paper studies one small exception to the GRS picture: a non-GRS $[6, 3, 4]_{11}$ MDS code whose projective deep-hole locus is exactly the $\mathbb{F}_{11}$-point set of a nonsingular conic. Its parity-check columns form the *Clebsch hexagon*, a six-arc in $\mathrm{PG}(2, 11)$. Thus a natural property of the covering-radius structure, with no assumed symmetry or conic construction, singles out a classical configuration.

The first main theorem starts from no symmetry or conic construction. Among all six-arcs of $\mathrm{PG}(2, 11)$, conic containment of the projective deep-hole syndrome locus reconstructs the Clebsch hexagon and its A_5 stabilizer. A line bound, a chord-defect identity, and Dye's Brianchon theorem give the conceptual implication; the census is needed only for the numerical clause. We prove quantitative gaps, reconstruct the Brianchon geometry and an invariant support bipartition from decoder ambiguity, and isolate $q = 11$.

The second main theorem asks what remains after passing from the parent to the full conic child. The unmarked conic code forgets its secant matching. The lost data reappears in the conic ideal: differences of matching-secant products, divided by the conic equation, form a finite quotient configuration. Its first two moments recover the unique unordered pair of factorization sheets, its signed cubic orients that pair, and a six-profile double-coset compression recovers a decorated parent from either singleton profile. Here *orientation* means a choice in a two-element C_2-torsor of sheets. It is distinct from the intrinsic unordered support bipartition recovered by decoder ambiguity, and has no implied quantum-information or chiral-polytope meaning.

The reflection-arrangement model places this reconstruction in the complete rank-three family. For a Clebsch hexagon K, the chord identity gives $|\mathcal{U}(K)| = q^2 - 14q + 45$, with off-conic excess $(q-4)(q-11)$ when $q \equiv 3 \pmod 4$. The fifteen chords are reduced projectivized H_3 mirrors. The six joins of a four-frame are similarly projectivized A_3 mirrors, and B_3 supplies the intermediate root-length-defect case. At $q = h+1 = 5, 7, 11$ the three reflection complements are full rational conics.

The configuration goes back to Clebsch [5, p. 336]. Edge constructed its finite-plane form and determined its order-60 stabilizer [6, §§29–32]; Dye proved projective uniqueness and determined the stabilizer over a general field [15, Theorems 1 and 3, pp. 275–278];

Storme and Van Maldeghem identified the corresponding six-arc in their finite-plane classification [23]. Deep holes of Reed–Solomon codes have both an algorithmic history [16, 9] and a geometric classification in the projective case [24]. Those results concern GRS codes, whose arcs lie on normal rational curves; the code here is not GRS.

What is new and what is not. The hexagon, its uniqueness and stabilizer, and its complete-exterior-set interpretation are classical [5, 6, 15, 2, 23]. Sadeh classified the six-arcs of $\mathrm{PG}(2, 11)$ [21]; we claim no priority for that census or its extension counts, and our frame-normalized replay is independent of the thesis. The companion arcs paper records the single conic-locus identification of Proposition 4.2, with a kernel-checked certificate [8, Prop. 8.2]; we reprove it for self-containment, and no result below depends on the companion paper. The paper-specific contribution begins with the symmetry-free coding criterion and its consequences: low-degree rigidity, quantitative gaps, the invariant support bipartition, decoder reconstruction, and the finite-field boundary. Edge already obtains the real Clebsch hexagons from the six opposite-vertex axes of an icosahedron [6, §§19–20]; the icosahedral mirror arrangement and its $6_5, 10_3, 15_2$ singularity ledger are likewise classical [4]. We claim no priority for this icosahedral geometry or its H_3 name. The finite-field characteristic-polynomial method is classical [1], and Jurrius and Pellikaan already derive coset-leader and list-weight enumerators from the intersection lattice of the parity-check derived arrangement, explicitly including redundancy-three MDS secant arrangements [19, Prop. 3.11, Thms. 5.3 and 5.7, Ex. 5.10]. Further paper-specific additions are the explicit $\mathrm{PGL}_3(\mathbb{F}_{11})$ bridge from the reduced H_3 fivefold points to the displayed code, its specialization to this code's decoder strata, the joint $A_3/B_3/H_3$ conic phase, and the composition of conic-ideal division, balanced recovery, cubic orientation, depth compression, and parent reconstruction. Raw one-factorizations, the $5, 14, 22$ marker fibres, matching-design interpretations, and coarse Hadamard orbitals are classical inputs.

2 A guided tour of forgetting and memory

The reconstruction problem begins with a tension. The projective deep-hole locus of the Clebsch code is exceptionally rigid: its containment in a conic determines the Clebsch class. The conic itself is familiar and forgetful: its twelve rational points support the extended Reed–Solomon code, with no preferred pairing, parent, or orientation. Figure 1 records how the missing information returns.

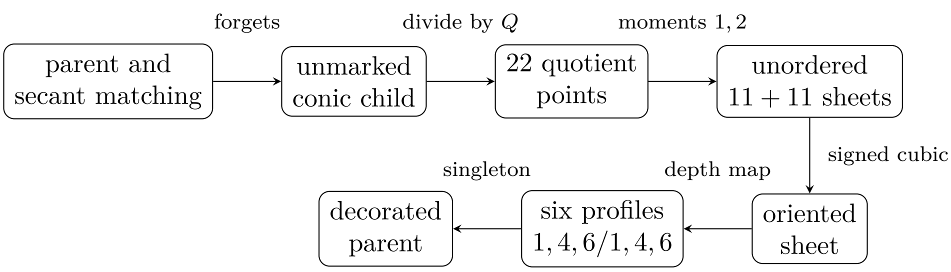

Figure 1: The reconstruction arc. The arrows are respectively a forgetting map, an ideal-quotient construction, intrinsic recovery of an unordered partition, a torsor orientation, a profile map, and decorated reconstruction; they are not all linear quotients.

For a perfect matching M of the conic points, let P_M be the product of its secant equations under the canonical bracket scaling of Section 8. All P_M have the same restriction to the conic, so $(P_M - P_N)/Q$ is a polynomial. The Coxeter-selected matchings give the finite quotient configurations used below.

Theorem 2.1 (Rigidity and the rank-three conic phase)**.** *The following statements hold.*

(A1) *Among six-arcs of* $\mathrm{PG}(2,11)$*, conic containment of the full projective deep-hole locus characterizes the Clebsch class and recovers its* A_5 *stabilizer.*

(A2) *For* A_3, B_3, H_3*, with Coxeter number* h*, the reflection-complement code has length*

$$(q-h/2)(q-h+1)$$

and minimum distance $(q-h/2-1)(q-h+1)$*; a nonmirror line meets the complement in at most* $q-h+1$ *points.*

(A3) *At* $q = h+1$ *the complement is the full rational conic and its projective-system code is extended GRS.*

(A4) *The square of a Coxeter element generates the split maximal torus in* $\mathrm{PSL}_2(q)$*, and its two moving point-orbits are the two Legendre cosets.*

Theorem 2.2 (Forgetting and balanced recovery)**.** (B1) *All canonically scaled matching-secant products have equal restriction to the conic; every four-endpoint switch difference is divisible by* Q*.*

(B2) *The* A_3, B_3, H_3 *quotient configurations have linear ranks* $3, 6, 10$*, respectively.*

(B3) *In the* B_3 *and* H_3 *configurations, the two Coxeter factorization sheets are, up to interchange, the unique complementary halves having equal first and second tensor moments.*

Theorem 2.3 (Cubic orientation and depth)**.** (C1) *With opposite signs on the two sheets, every even signed profile moment vanishes, and the primitive weighted barycentre relation* $v_1 + 4v_4 + 6v_6 = 0$ *kills degree one.*

(C2) *Within the signed tensor-moment family and its linear functionals, degree three is the first surviving degree. Its nonzero tensor transforms by the outer character and has kernel stabilizer.*

(C3) *For* $G = \mathrm{PGL}_2(11)$*,* $H = A_5$*, and the common* $K = A_4$*, the space* $K\backslash G/H$ *has six labels in two triples of sizes* $1, 4, 6$*, with profiles* $v_1, v_4, v_6, -v_1, -v_4, -v_6$*.*

(C4) *The profile map has rank two and four-dimensional linear kernel, while separating all six labels as a set.*

(C5) *The singleton profiles recover the unordered golden matching pair; a choice of sheet and its singleton decoration recovers a Clebsch parent.*

(C6) *In characteristic eleven the depth plane is* $P(1)^{A_4}/\operatorname{soc} P(1)$*, where the projective cover* $P(1)$ *has Loewy dimensions* $1 \mid 9 \mid 1$*. The three positive profile rays intrinsically recover the orbit sizes* $1, 4, 6$ *and stabilizer orders* $12, 3, 2$*.*

Theorem 2.4 (Rank-three arithmetic gluing)**.** (D1) *The* A_3 *matching datum fuses. The* B_3 *and* H_3 *data split into two silver and two golden sheets, respectively.*

(D2) *In the* H_3 *case the two displayed* A_5 *stabilizers meet in* A_4*, generate* $\mathrm{PSL}_2(11)$*, and are exchanged by an outer projectivity carried by the rational* S_4/A_4 *hinge.*

(D3) *The fixed conic child has a twenty-two-parent fibre whose intrinsic two-point quotient is the same free orientation torsor detected by the sheet character; choosing a point is exactly one bit of decoration.*

The sequence $22 \to 6 \to 2 \to 1$ is an information ledger, not a chain of faithful linear images. In particular, the two-dimensional profile plane is not the two-element sheet set. Sections 8–12 prove the new arrows after the rigidity and Coxeter inputs have been established.

How the proofs divide. The headline arguments isolate mechanisms: the Plücker relation remembers factorization modulo the conic ideal; an annihilator line forces the balanced sheets; antipodality and the primitive $1:4:6$ relation force cubic-first orientation; projective-cover structure explains the depth plane; and determinant-sign descent identifies the surviving bit. Exact finite computations verify the frozen orbit, rank, and normalizer hypotheses used by these arguments. They are supporting certificates; the preceding structures explain the headline theorems.

Independence and extension. The unlabelled child admits no equivariant selector of a parent, whereas a marked matching supplies exactly the one bit needed to make the selection. In Hofstadter's terms [17, "Contracrostipunctus"], the unlabelled child is a record that no equivariant record player can play: decorations are styli, and the lattice $22 \to 6 \to 2 \to 1$ lets different styli play different tracks. This is independence from the unlabelled theory and decidability in a one-bit extension, not undecidability.

3 The Clebsch hexagon

Fix the nonsingular conic

$$\mathcal{C}: \ XZ = Y^2$$

in $\mathrm{PG}(2,11)$, with its 12 points identified with $\mathrm{PG}(1,11) = \mathbb{F}_{11} \cup \{\infty\}$ in the usual way. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}(3,11)$ is $\mathrm{PGL}_2(11)$, acting on $\mathcal{C}$ as it acts on the projective line.

A convenient representative is

$$A = \{(1,10,0),\ (1,9,1),\ (1,4,7),\ (1,8,5),\ (0,1,4),\ (1,1,7)\}.$$

It is a six-arc disjoint from $\mathcal{C}$; its fifteen joins also avoid $\mathcal{C}$. The following invariant construction explains its symmetry.

Definition 3.1 (Clebsch hexagon)**.** Let $A_5 \subset \mathrm{PGL}_2(11)$ be an icosahedral subgroup. The six Sylow 5-subgroups of A_5 each fix two points of $\mathcal{C}$. The *Clebsch hexagon* is the set of poles of the six chords joining these fixed pairs.

For a nonsingular conic over an odd field, an *external point* lies on two tangents and an *internal point* on none; an external, or *passant*, line is disjoint from the conic. Edge's six vertices are external, their fifteen joins are external, and those joins concur three at a time at ten internal Brianchon points [6, §§29–32]. Thus the hexagon is a complete exterior set in the terminology of Blokhuis, Seress, and Wilbrink [2, pp. 143,146]. Their $q = 11$ list also contains a Pasch configuration, but its four collinear triples prevent it from defining an MDS code. Thus exteriority alone gives only $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$; the arc hypothesis selects the Clebsch branch, and Proposition 4.2 proves equality.

This is the arc K_2 of Storme and Van Maldeghem [23, Props. 11–12] and Dye's synthetic hexagon when 5 is a square in the base field [15, Theorem 1]. Dye proves uniqueness, stabilizer A_5, and the unique associated polarity; in the displayed coordinates its conic is $\mathcal{C}$. His calculation also shows that the six vertices lie on no conic in characteristic 11 [15, Theorem 2, pp. 277–278; p. 281]. Storme and Van Maldeghem record that the arc is incomplete at $q = 11$ [23, Prop. 13], so the associated redundancy-three code has covering radius three.

4 The code and its projective syndrome locus

Form the 3×6 matrix H whose columns are (representatives of) the six points of A, and let

$$C = \{c \in \mathbb{F}_{11}^6 : Hc^{\mathsf{T}} = 0\}$$

be the resulting $[6, 3, 4]_{11}$ MDS code. Since the six columns of H do not lie on any conic (their evaluation matrix against the six quadratic monomials $X^2, XY, XZ, Y^2, YZ, Z^2$ is nonsingular), C is not monomially equivalent to a GRS code: the dual of a GRS code is GRS, and the parity-check column system of a length-six, dimension-three GRS code lies on a normal rational curve, here a conic, in the classical normal-rational-curve/GRS dictionary [22].

The arc–coset dictionary from the introduction applies without a GRS hypothesis [10, Def. 6.1, Thm. 6.3]. Each uncovered direction x lifts to $q - 1$ weight-three cosets, and each has $\binom{6}{3} = 20$ minimum-weight leaders. Geometrically, the same condition says that adjoining x as a seventh column preserves the MDS property.

The stabilizer already forces the small point sets that will occur.

Proposition 4.1 (The A_5 point orbits)**.** *Let $G = \mathrm{Stab}(A) \cong A_5$. Its orbits on* $\mathrm{PG}(2, 11)$ *have lengths*

$$6,\ 10,\ 12,\ 15,\ 30,\ 30,\ 30.$$

The first four are respectively A, the ten Brianchon points, $\mathcal{C}(\mathbb{F}_{11})$, and the fifteen vertices of the five self-polar triangles. In particular, $\mathcal{C}(\mathbb{F}_{11})$ is the unique twelve-point G-orbit.

Proof. Dye's polarity and stabilizer theorems identify G with the ternary icosahedral representation preserving $\mathcal{C}$, and his Corollary 1 makes it transitive on the six vertices, ten Brianchon points, and fifteen triangle vertices [15, Theorems 2–3 and Corollary 1, pp. 276–279]. We derive the orbit sizes without using the finite enumeration.

The argument first determines the points with each possible nontrivial stabilizer, then applies orbit–stabilizer to recover the orbit lengths.

Since $11 \nmid 60$, the representation is semisimple, and the ternary icosahedral representation is irreducible; hence G fixes no projective point. The ordinary three-dimensional icosahedral character, reduced in $\mathbb{F}_{11}$, shows that an involution, an element of order three, and an element of order five fix respectively 13, 1, and 3 projective points. Indeed, an involution has a one-dimensional eigenspace and a two-dimensional eigenspace; the nontrivial order-three eigenvalues are not in $\mathbb{F}_{11}$; and the three order-five eigenvalues are distinct because $5 \mid 10$. Restricting the same representation to the standard subgroups of A_5 gives the following common-fixed-point ledger; the standard subgroup classification of A_5 shows that these are all possible nontrivial proper point stabilizers. Here D_5 has order ten, and the last row counts points whose stabilizer is exactly the displayed subgroup type. For the noncyclic entries, determine the common projective eigenlines of standard generators: a D_5 normalizer retains one of the three C_5 eigenlines, an S_3 normalizer retains the unique C_3 eigenline, a V_4 has three simultaneous eigenlines, and its overgroup A_4 permutes those three and fixes none. This gives the second row.

H	A_4	D_5	S_3	C_5	V_4	C_3
number of conjugate subgroups	5	6	10	6	5	10
fixed points for one H	0	1	1	3	3	1
points with stabilizer conjugate to H	0	6	10	12	15	0

By orbit–stabilizer, the third row already gives the orbits of lengths 6, 10, 12, and 15; only points with exact stabilizer C_2 remain. To spell out the subtraction in the last row: the normalizer D_5 fixes one of the three C_5 eigenlines and interchanges the other two, giving 6 points with stabilizer D_5 and 12 with stabilizer C_5; the unique C_3-fixed line is already fixed by its S_3 normalizer; and A_4 cycles the three V_4 eigenlines, leaving all fifteen with exact stabilizer V_4.

It remains to account for exact C_2 stabilizers. Each involution lies in two D_5 subgroups, two S_3 subgroups, and one V_4. Of its thirteen fixed points, these contribute respectively 2, 2, and 3 points from the preceding rows. The remaining six points have exact stabilizer C_2. There are fifteen involutions, so this gives 90 points, necessarily three orbits of length $60/2 = 30$. The total $6 + 10 + 12 + 15 + 90 = 133$ exhausts the plane. Dye's Theorem 6(v), specialized at $q = 11 \equiv 1 \pmod{10}$, identifies the twelve-point C_5-stabilizer orbit with $\mathcal{C}(\mathbb{F}_{11})$ [15, pp. 281–282; §3.1, p. 284]; the other three identifications are the transitive sets cited above. □

Proposition 4.2 (The uncovered locus is the conic)**.** *The covering radius of C is three, and*

$$\mathcal{U}(A) = \mathcal{D}_{\text{proj}}(C) = \mathcal{C}(\mathbb{F}_{11}).$$

Proof. The set $\mathcal{U}(A)$ is G-invariant. Dye's Clebsch classification gives exactly ten Brianchon points. The relevant double count is short. The fifteen edges contribute 150 off-arc incidences. The 45 pairs of disjoint edges meet once, so if n_i counts off-arc points lying on exactly i edges, then a point on three edges accounts for $\binom{3}{2} = 3$ such pairs and

$$n_1 + 2n_2 + 3n_3 = 150, \qquad n_2 + 3n_3 = 45, \qquad n_3 = 10.$$

Thus the edges cover $n_1 + n_2 + n_3 = 115$ of the 127 off-arc points, leaving $|\mathcal{U}(A)| = 12$. This is the $q = 11$ specialization of the chord-defect identity in Lemma 7.1. By definition $\mathcal{U}(A)$ is disjoint from the unique six-point orbit A. The orbit lengths in Proposition 4.1 therefore force this invariant twelve-set to be the unique twelve-point orbit $\mathcal{C}(\mathbb{F}_{11})$. By the dictionary above this is exactly the projective deep-hole syndrome locus, and its nonemptiness gives covering radius three rather than two. Dye's edge criterion and the complete-exterior-set description of Blokhuis–Seress–Wilbrink independently supply the classical inclusion $\mathcal{C}(\mathbb{F}_{11}) \subseteq \mathcal{U}(A)$ [15, discussion preceding Theorem 6, p. 281]; see also [2, pp. 143, 146]. □

Corollary 4.3 (Directions, cosets, leaders, and received words)**.** *$\mathcal{U}(A)$ has 12 directions, exactly the points of $\mathcal{C}$. Above them lie $12(11 - 1) = 120$ nonzero maximum-distance syndromes, hence 120 deep-hole cosets. Each such coset has $\binom{6}{3} = 20$ minimum-weight leaders, for 2400 leaders in all, and contains $|C| = 11^3 = 1331$ received words. Consequently C has $120 \cdot 1331 = 159720$ received-word deep holes.*

Proof. Immediate from Proposition 4.2, the leader-count clause of the DMP dictionary, and the fact that the fiber of $\mathbb{F}_{11}^3 \setminus \{0\} \to \mathrm{PG}(2, 11)$ over each direction consists of ten nonzero syndromes, hence ten distinct cosets. Every coset has $|C|$ words. □

4.1 Automorphisms and deep-hole orbits

The projective stabilizer of the six column rays is A_5 of order 60, acting faithfully and 2-transitively on the coordinate supports. It is the support-permutation quotient of the monomial automorphism group:

$$1 \longrightarrow \mathbb{F}_{11}^{\times} \longrightarrow \mathrm{MAut}(C) \longrightarrow A_5 \longrightarrow 1.$$

The kernel consists of the ten global coordinate scalars. The sequence splits. Since $\gcd(3, 10) = 1$, each projective class has a unique determinant-one representative; their monomial lifts give an A_5 complement. Hence

$$\mathrm{MAut}(C) \cong C_{10} \times A_5$$

has order 600. For the displayed column representatives, the subgroup of pure coordinate permutations is trivial.

Proposition 4.4 (One orbit on received-word deep holes)**.** *The monomial automorphism group of C is transitive on the 120 deep-hole cosets. Let Γ be the group generated by these automorphisms and by translations $v \mapsto v + c$ for $c \in C$. Then*

$$|\Gamma| = |C|\,|\mathrm{MAut}(C)| = 1331 \cdot 600 = 798600,$$

and Γ is transitive on all 159720 received-word deep holes. The stabilizer of each such word is cyclic of order five.

Proof. The projective stabilizer A_5 is transitive on the twelve points of $\mathcal{C}$. Indeed, its six Sylow 5-subgroups form one conjugacy class, each fixes the two endpoints of one of the six fixed-point chords, and the involution in its dihedral normalizer interchanges those endpoints. These twelve endpoints exhaust $\mathcal{C}(\mathbb{F}_{11})$. Every such projectivity lifts to a monomial automorphism of the parity-check columns, so the monomial group is transitive on the twelve syndrome rays above $\mathcal{C}$. Its central scalar subgroup $\mathbb{F}_{11}^{\times}$ acts transitively on the ten nonzero vectors of each ray. Hence the monomial group is transitive on all $12 \cdot 10 = 120$ maximum-distance syndromes and therefore on the corresponding cosets. Given deep holes v and w, choose $m \in \mathrm{MAut}(C)$ with $\sigma(mv) = \sigma(w)$. Then $c = w - mv$ has zero syndrome, so $c \in C$, and the codeword translation by c sends mv to w. Both kinds of generator are Hamming isometries preserving C. The monomial group normalizes the translation subgroup, since $m\, t_c\, m^{-1} = t_{m(c)}$ with $m(c) \in C$. Thus $\Gamma = C \rtimes \mathrm{MAut}(C)$; the two subgroups have trivial intersection and the displayed product order follows. Orbit–stabilizer now gives stabilizer order $798600/159720 = 5$, and every group of prime order is cyclic. $\square$

4.2 Decoding consequences

The coset-leader-weight distribution of C is $(1, 60, 1150, 120)$ (cosets of weight $0, 1, 2, 3$, summing to $11^3 = 1331$); the last entry counts deep-hole cosets, not received words. Its codeword weight enumerator is $(1, 0, 0, 0, 150, 420, 760)$, forced by the MDS parameters.

The special geometry also makes distance diagnosis and nearest-word structure unusually explicit. Write a syndrome as $s = (s_0, s_1, s_2)$ and put $Q(s) = s_0 s_2 - s_1^2$. For a point $x \notin A$, its *secant index* is the number of chords of A through x; by the dictionary above, this is the number of weight-two leaders for each nonzero syndrome on the ray x.

Proposition 4.5 (Complete syndrome oracle and ambiguity enumerator)**.** *For $v \in \mathbb{F}_{11}^6$ with syndrome $s = Hv^{\mathsf{T}}$,*

$$d(v,C) = \begin{cases} 0, & s = 0, \\ 1, & [s] \in A, \\ 3, & s \neq 0 \text{ and } Q(s) = 0, \\ 2, & \text{otherwise.} \end{cases}$$

Among the 1330 *nonzero cosets, the numbers having respectively one, two, three, and twenty nearest codewords are*

$$960, \qquad 150, \qquad 100, \qquad 120.$$

For a distance-two syndrome the number of nearest codewords is the secant index of its direction. Every distance-three syndrome has exactly one leader on each of the twenty three-element coordinate supports.

Proof. The DMP dictionary identifies weight-one directions with the six columns, weight at most two with the secant union, and weight three with its complement. Proposition 4.2 identifies the last set with $Q = 0$, proving the oracle. In the notation of its proof, the chord-intersection equations give $(n_0, n_1, n_2, n_3) = (12, 90, 15, 10)$, later recognized as the H_3 strata in Section 7.1, where n_i counts projective directions of secant index i and n_0 is the uncovered locus. Every direction has ten nonzero affine syndromes. Thus the distance-two cosets contribute 900 cases of multiplicity one, 150 of multiplicity two, and 100 of multiplicity three; the sixty weight-one cosets add sixty more cases of multiplicity one. Leaders are in bijection with nearest codewords by translation within the coset. Finally, the three columns on any support are independent. For an uncovered syndrome their unique coefficients are all nonzero, since a zero coefficient would put the syndrome on a secant. Hence all twenty supports, and no others, give its weight-three leaders. □

4.3 Self-polar structure

Edge's finite-plane construction [6, §§30.2–31] and Dye's general-field synthetic theory [15, Theorem 2, pp. 277–278] exhibit five triangles on the six vertices, self-polar for $\mathcal{C}$. Their vertex-pair matchings partition the fifteen pairs into a *synthematic total*, that is, a partition of the edges of K_6 into five perfect matchings. Section 6 turns its ten pairs of matchings into the Brianchon–support dictionary.

5 The rigidity theorem

Proposition 4.2 concerns one known arc. The following theorem is the paper's main claim: among *all* six-arcs of $\mathrm{PG}(2,11)$, the property "the uncovered locus lies on a conic" picks out the Clebsch hexagon uniquely, and does so in a way that recovers its projective stabilizer from the coding condition rather than assuming it.

Remark 5.1 (Two conditions, one word apart)*.* The condition studied here is that $\mathcal{U}(A)$ — the projective uncovered, or projective deep-hole syndrome, locus — lies on a conic. It is *not* the familiar condition that the arc A itself lies on a conic, which is the standing hypothesis throughout the arc literature and which the Clebsch hexagon famously fails. The two are logically unrelated: they constrain different point sets, and the arcs satisfying them are disjoint. Indeed every six-arc lying *on* a conic in $\mathrm{PG}(2,11)$ has $|\mathcal{U}(A)| \in \{18, 19, 20, 22\}$, so clause (iii) of Theorem 5.3 below excludes all of them. This four-value assertion is the conic-inscribed subcensus of the independent rigidity replay listed in Table 3.

We first isolate the geometric input that removes degenerate conics from the computer-assisted part of the argument.

The external-line case is a six-point instance of the nonexistence in odd order of nontrivial *hyperfocused arcs*, whose secants determine only $k-1$ points on an external line [3, 7, 20]. We give a direct proof because the three possible line positions are then handled together.

Lemma 5.2 (Six-arc line bound)**.** *Let A be a six-arc in* $\mathrm{PG}(2,q)$*, where q is odd. Every line contains at most $q-5$ points of $\mathcal{U}(A)$.*

Proof. A chord contains no point of $\mathcal{U}(A)$. If a non-chord line ℓ contains one vertex, the ten chords on the other five vertices meet ℓ in at least five distinct points. Indeed, two of those chords sharing an endpoint meet only at that endpoint, which lies off ℓ; their intersections with ℓ therefore give a proper edge-colouring of K_5, whose chromatic index is five. Together with the vertex, at least six of the $q+1$ points of ℓ are therefore outside $\mathcal{U}(A)$.

It remains to consider $\ell \cap A = \varnothing$. The fifteen chords meet ℓ in covered points, and at most three chords pass through any one of them: chords sharing an endpoint meet only at that arc vertex, not on ℓ, so all chords through one point of ℓ have disjoint endpoint pairs. Hence there are at least five covered points. Equality would give a proper five-edge-colouring of K_6, necessarily a one-factorization. After relabelling, three factors are

$$01|23|45, \qquad 02|14|35, \qquad 03|15|24.$$

Send ℓ to infinity and normalize the first two directions to horizontal and vertical. Put $p_0 = (0,0)$, $p_1 = (x,0)$ and $p_2 = (0,y)$, where $xy \neq 0$. Writing $p_3 = (a,y)$, the first two factors force $p_4 = (x,b)$ and $p_5 = (a,b)$. Parallelism of 03 with 15 and with 24 gives two determinant equations whose difference is $2xy = 0$, impossible for odd q. Thus a disjoint line has at least six covered points, and in all cases at most $(q+1)-6 = q-5$ points of $\mathcal{U}(A)$ remain. □

Theorem 5.3 (Rigidity theorem)**.** *For a six-arc $A \subset \mathrm{PG}(2,11)$, the following are equivalent:*

(i) the uncovered locus $\mathcal{U}(A)$ lies on some conic (possibly degenerate);

(ii) $\mathcal{U}(A)$ is exactly the set of $\mathbb{F}_{11}$-points of a nonsingular conic;

(iii) $|\mathcal{U}(A)| \leq 15$ (in fact, $|\mathcal{U}(A)| = 12$);

(iv) A is $\mathrm{PGL}(3,11)$*-equivalent to the Clebsch hexagon;*

(v) the stabilizer of A in $\mathrm{PGL}(3,11)$ *contains A_5.*

Proof. Suppose first that $\mathcal{U}(A)$ lies on a conic Q. If Q is nonsingular then $|\mathcal{U}(A)| \leq q+1 = 12$. If Q is degenerate, its rational points lie on one or two rational lines when it splits; in the nonsplit rank-two case its only rational point is the singular point. Thus Lemma 5.2 again gives $|\mathcal{U}(A)| \leq 12$. On the other hand, the chord-defect identity of Lemma 7.1 says

$$|\mathcal{U}(A)| = 22 - c(A),$$

where $c(A)$ is the number of nonvertex triple-edge concurrences. In Dye's terminology these are precisely the Brianchon points of the hexagon A. Over a field of characteristic different from two, Dye proves that a hexagon has at most ten Brianchon points and that the equality cases form a single $\mathrm{PGL}_3(K)$-orbit [15, §2.2–2.3, Theorem 1, pp. 275–276].

Hence $c(A) \leq 10$, so $|\mathcal{U}(A)| \geq 12$; equality holds throughout, $c(A) = 10$, and A is a Clebsch hexagon. This proves (i)⇒(iv). Proposition 4.2 and projective equivariance give (iv)⇒(ii)⇒(i). Dye's stabilizer theorem gives (iv)⇒(v), and Storme and Van Maldeghem's finite-plane classification gives the converse.

It remains only to connect the numerical condition (iii). Any ordered four points of a six-arc form a projective frame, uniquely normalized to $(1,0,0), (0,1,0), (0,0,1), (1,1,1)$. Enumerating the remaining unordered pair subject to the arc condition gives 1548 normalized sets. Each embedded arc has $\binom{6}{4} 4! = 360$ ordered-frame choices. For each choice, apply the unique normalizing projectivity, sort the resulting six normalized points, and encode that list lexicographically. Taking the least of the 360 lists removes the frame choice; two arcs have the same least list exactly when a projectivity carries one to the other. This canonical key gives fifteen classes, and the number of distinct normalized presentations of a class with stabilizer G is $360/|G|$. The resulting extension counts are

$$|\mathcal{U}(A)| \in \{12, 16, 18, 19, 20, 21, 22\}.$$

Several of the fifteen projective classes share an extension count. Across the 1548 frame-normalized representatives, the corresponding multiplicities are

$$(6, 30, 150, 300, 630, 360, 72).$$

The six representatives attaining 12 form the single Clebsch class; every other class has at least 16 extension points. Thus (iii)⇔(iv), completing the equivalence. The exhaustive calculation is therefore used for the size-gap clause, not for the conic containment implication. □

The same census supports a strengthening in which a cubic, rather than a conic, is allowed.

Proposition 5.4 (Low-degree algebraic rigidity)**.** *For a six-arc $A \subset \mathrm{PG}(2, 11)$, the uncovered locus $\mathcal{U}(A)$ is contained in the zero set of a nonzero homogeneous form of degree at most three if and only if A is projectively equivalent to the Clebsch hexagon. In the Clebsch case the least such degree is two: the quadratic vanishing space is spanned by the defining nonsingular conic equation Q, and the cubic vanishing space is*

$$Q \cdot \mathbb{F}_{11}[X, Y, Z]_1.$$

Computer-assisted proof. For each of the fifteen projective classes in the six-arc census, evaluate the ten cubic monomials on $\mathcal{U}(A)$. The resulting evaluation map has rank ten for every non-Clebsch class and rank seven for the Clebsch class. Thus no nonzero cubic vanishes on a non-Clebsch uncovered locus. A containing form of smaller positive degree could be multiplied by a nonzero monomial to produce such a cubic, so smaller positive degrees are excluded as well; no nonzero constant vanishes on the nonempty locus. For the Clebsch class, the quadratic kernel is generated by Q, and multiplication by linear forms gives the full three-dimensional cubic kernel. □

Corollary 5.5 (Monomial characterization)**.** *Up to monomial equivalence, C is the unique linear $[6, 3, 4]_{11}$ code of covering radius three whose projective deep-hole syndrome locus is contained in a plane curve of degree at most three. In particular, A_5 is recovered from this coding condition, not assumed as a hypothesis.*

Proof. The columns of a parity-check matrix of such a code form a six-arc A, and covering radius three identifies its projective deep-hole syndrome locus with $\mathcal{U}(A)$. Proposition 5.4

makes A projectively equivalent to the Clebsch hexagon. A projectivity between two unordered parity-check column sets lifts, after choosing column representatives, to a row operation together with a coordinate permutation and nonzero coordinate scalings; these are exactly a monomial code equivalence. The converse is immediate, and the stabilizer statement is clause (v) of the theorem. □

Remark 5.6 (Sharpness of the degree threshold). The cutoff cannot be raised from three to four: one non-Clebsch class has an 18-point uncovered locus equal to the rational zero set of a quartic. It is normalized class `C02` in the low-degree replay of Table 3, which prints the exact equation and verifies the equality point by point. No smoothness, irreducibility, or genus assertion is used here.

The rigidity theorem has quantitative consequences at three scales: the number of deep-hole directions, the distance of their locus from the nearest conic, and the effect of replacing one vertex of the hexagon.

For the second, let $\mathfrak{C}$ be the set of nonsingular conics in $\mathrm{PG}(2,11)$ and define the *nearest-conic discrepancy*

$$\delta(A) = \min_{\mathcal{Q}\in\mathfrak{C}} |\mathcal{U}(A) \triangle \mathcal{Q}(\mathbb{F}_{11})|.$$

This is a PGL-invariant function, not a metric on arcs: equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A)$, while PGL permutes $\mathfrak{C}$.

For the local statement, call two embedded six-arcs *neighbors* when one is obtained from the other by replacing one point. Fix the displayed Clebsch arc A and its standard conic $\mathcal{C}$, and put

$$d_{\mathcal{C}}(B) = |\mathcal{U}(B) \triangle \mathcal{C}|$$

for every embedded six-arc B.

Theorem 5.7 (Quantitative gaps). *(a) There is no six-arc $A \subset \mathrm{PG}(2,11)$ with $12 < |\mathcal{U}(A)| < 16$. Every arc attaining $|\mathcal{U}(A)| = 12$ is Clebsch up to projectivity. If $|\mathcal{U}(A)| \geq 16$, then $\mathcal{U}(A)$ lies on no conic.*

(b) Every non-Clebsch class satisfies $\delta(A) \geq 12$, and the bound is sharp.

(c) The arc A has 252 distinct neighbors. Every neighbor B satisfies $d_{\mathcal{C}}(B) \geq 18$ and $|\mathcal{U}(B) \cap \mathcal{C}| \leq 7$; none has its uncovered locus contained in a conic, degenerate or otherwise.

The first distribution below counts projective classes. The next two count embedded one-point replacements of the fixed arc A; the final list then groups those replacements into A_5-orbits:

$$\begin{aligned} \#\{[B] : \delta(B) = d\} &= \{0{:}1, 12{:}2, 13{:}1, 14{:}3, 15{:}1, 16{:}4, 17{:}2, 18{:}1\}, \\ \#\{B \sim A : d_{\mathcal{C}}(B) = d\} &= \{18{:}30, 19{:}60, 20{:}90, 22{:}42, 24{:}30\}, \\ \#\{B \sim A : |\mathcal{U}(B) \cap \mathcal{C}| = m\} &= \{4{:}60, 6{:}132, 7{:}60\}. \end{aligned} \tag{1}$$

Under A_5, the neighbors split into eight orbits with (orbit size, discrepancy)

$$(30, 18), (60, 19), (30, 20), (30, 20), (30, 20), (12, 22), (30, 22), (30, 24).$$

Remark 5.8 (Global versus fixed-conic discrepancy). The local bound is tied to the fixed pair $(A, \mathcal{C})$. Indeed, the conic-preserving projectivity $g : (X, Y, Z) \mapsto (4X, 2Y, Z)$ sends A to a distinct embedded six-arc, while equivariance gives $\mathcal{U}(gA) = g\mathcal{U}(A) = \mathcal{C}$. Hence $d_{\mathcal{C}}(gA) = 0$ although $gA \neq A$. The global statement therefore minimizes over conics and passes to projective classes; the bound eighteen applies only to one-point replacements of the fixed arc.

Computer-assisted proof. The first clause restates Theorem 5.3. For the second, the enumeration normalizes all 1548 representatives to fifteen projective classes, enumerates the 160930 nonsingular conics, and computes the first distribution in (1).

For the third, it deletes each vertex in turn, tests every replacement, and obtains 42 choices per vertex and 252 distinct neighbors. Exact quadratic-monomial evaluation matrices have full rank six, so no nonzero quadratic vanishes on a neighbor's uncovered locus. The remaining two distributions and the eight A_5-orbits follow from the sixty stabilizing projectivities. ☐

Remark 5.9 (Why this rigidity is not in the extension-locus literature). The set $\mathcal{U}(A)$ is the classical extension locus, studied for sufficiently large arcs through Segre's tangent-envelope method; see Ball and Lavrauw [11, Thms. 39–41]. Those results require arc-size hypotheses far from the six-point regime, while Sadeh's census uses six-arcs for a different classification problem. Neither direction records the conic rigidity detected here.

6 The invariant support bipartition

Label the six coordinates by $0, \ldots, 5$. The 20 three-element coordinate supports form two complementary A_5-orbits of size ten. Automorphisms preserve the two classes, but nothing intrinsic chooses one as positive. We use *support chirality* only as shorthand for this unoriented bipartition. Its ten complementary pairs carry the Petersen graph in Figure 2.

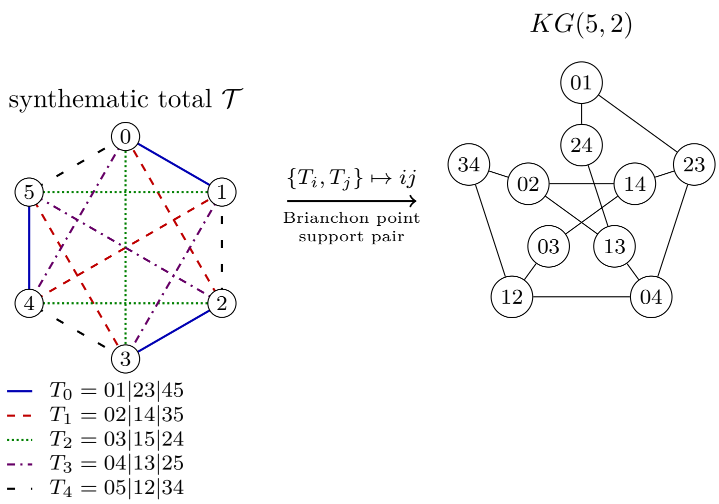

Figure 2: The synthematic–Petersen dictionary. The five line styles encode the five self-polar triangles as perfect matchings of the six coordinates. For each pair $\{T_i, T_j\}$, their union is an alternating six-cycle: its opposite-edge matching gives a Brianchon point, and its alternating vertex classes give a complementary support pair. Both are represented by the Petersen vertex ij; adjacency means that the index pairs are disjoint.

Proposition 6.1 (Brianchon–support dictionary). *Let $\mathcal{T}$ be the five self-polar triangles of Edge and Dye [6, 15], regarded as the five perfect matchings in their synthematic total, and let $\mathcal{B}(A)$ be the set of ten Brianchon points of the Clebsch hexagon. For distinct $T, T' \in \mathcal{T}$, the union $T \cup T'$ is an alternating six-cycle. Let $M(T, T')$ be the perfect matching joining opposite vertices of this cycle. Let $\beta(T, T') = \{S, S^{\mathsf{c}}\}$ be the unordered pair of alternating vertex classes of the cycle. For example, $T_0 \cup T_1$ is the cycle $0, 1, 4, 5, 3, 2$, so $M(T_0, T_1) = 05|13|24$ and $\beta(T_0, T_1) = \{\{0, 3, 4\}, \{1, 2, 5\}\}$.*

The three chords indexed by $M(T, T')$ concur at a point $b(T, T') \in \mathcal{B}(A)$. The maps

$$\binom{\mathcal{T}}{2} \longrightarrow \mathcal{B}(A), \qquad \{T, T'\} \longmapsto b(T, T'),$$

and

$$\binom{\mathcal{T}}{2} \longrightarrow \{\{S, S^{\mathrm{c}}\} : |S| = 3\}, \qquad \{T, T'\} \longmapsto \beta(T, T'),$$

are A_5-equivariant bijections. They identify the ten Brianchon points with the ten complementary support pairs. Under this identification, Petersen adjacency is disjointness of the corresponding two-element subsets of $\mathcal{T}$.

Proof. The opposite-vertex matching is disjoint from T and T' and contains one edge from each of the other three members of the synthematic total. Hence $\{T, T'\}$ is recovered as the two members of $\mathcal{T}$ disjoint from $M(T, T')$, so the ten triangle pairs give exactly the ten perfect matchings outside the total. Applying the alternating-class rule to the five displayed matchings gives ten distinct complementary support pairs, hence all ten such pairs. Edge's Clebsch construction says that the three chords of each such matching concur, at the ten distinct Brianchon points [6, §§19–20,30]. The bipartition bijection and Petersen assertion are the alternating-cycle construction above.

The ten outside matchings give the ten triple concurrences. The remaining fifteen intersections of disjoint chords have multiplicity one. Since every step uses only the invariant total and incidence, both bijections and their compatibility with the support map are A_5-equivariant. ☐

Corollary 6.2 (The decoder reconstructs the Brianchon configuration)**.** *The ten projective directions having three nearest weight-two errors are exactly the ten Brianchon points. At each such direction the three error supports are pairwise disjoint and hence form a perfect matching of the six coordinates. The ten matchings thereby recovered are precisely the ten perfect matchings outside the five-matching synthematic total.*

Proof. The three weight-two representations of a secant-index-three direction use the three chords through it. Proposition 6.1 identifies the ten triple concurrences with the ten Brianchon points and their chords with exactly the ten matchings complementary to the total. ☐

Proposition 6.3 (Invariant support bipartition)**.** *Only the bipartition is intrinsic: neither part has a canonical sign. The twenty three-element coordinate supports split into two complementary A_5-orbits $\mathcal{O}_+$ and $\mathcal{O}_-$ of size ten. For every maximum-distance syndrome s, each support determines exactly one weight-three leader of the coset $H^{-1}(s)$; hence that coset has ten leaders supported in each class. Globally, the 2400 minimum-weight leaders split as $1200 + 1200$.*

Every monomial automorphism of C preserves this unordered bipartition. The other coset in the exotic normalizer $N_{S_6}(A_5) \cong S_5$ exchanges the two support classes, but its sixty elements are not automorphisms of the displayed code. By Proposition 6.1, the ten complementary support pairs are canonically both the two-subsets of the five self-polar triangles and the ten Brianchon points.

Proof. The degree-six action of A_5 on the twenty three-subsets has two ten-element orbits, exchanged by complementation. It preserves a unique synthematic total, also preserved by its full S_5 normalizer. The alternating-cycle construction of Figure 2 is therefore equivariant, and the outside normalizer coset exchanges the two support orbits. Fix a maximum-distance syndrome s and a three-support S. The three columns indexed by S are independent because A is an arc, so they give a unique coefficient vector representing s. None of its three coefficients can vanish: otherwise $[s]$ would lie on a secant of A, contrary to maximum distance. The resulting error vector therefore has weight three and is the unique leader with support S. Finally, every monomial automorphism induces a support permutation in A_5, by the exact sequence in Section 4.1, and so preserves each support orbit. The outside normalizer coset swaps the two orbits but cannot preserve C, even after coordinate rescaling, because its support permutations lie outside that quotient A_5. □

7 Why $q = 11$

Call an arc A *conic-filling* if $\mathcal{U}(A) = \mathcal{Q}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{Q}$. The conic-filling syndrome phenomenon of Sections 4–5 is specific to $q = 11$, even without a symmetry hypothesis on the arc. A universal chord-multiplicity identity makes this a four-field question, and the only nontrivial residue is a classical exact-distance problem in the Sylvester graph.

Lemma 7.1 (Chord-defect identity)**.** *Let A be a six-arc in* $\mathrm{PG}(2, q)$*, and let $c(A)$ be the number of points off A incident with three of its fifteen chords. Then*

$$|\mathcal{U}(A)| = q^2 - 14q + 55 - c(A), \qquad 0 \le c(A) \le 15.$$

Consequently, if $|\mathcal{U}(A)| = q + 1$, then

$$c(A) = (q - 6)(q - 9).$$

Proof. At a point off A, concurrent chords have pairwise disjoint endpoint pairs, so at most three chords occur. Let n_i count the off-arc points on exactly i chords. Counting chord–point incidences and pairs of chords with disjoint endpoints gives

$$n_1 + 2n_2 + 3n_3 = 15(q - 1), \qquad n_2 + 3n_3 = \binom{15}{2} - 6\binom{5}{2} = 45.$$

Writing $c(A) = n_3$, the number of covered points off A is $n_1 + n_2 + n_3 = 15q - 60 + c(A)$. Subtracting this from the $q^2 + q - 5$ points off A proves the identity. Each triple concurrence uses one of the fifteen perfect matchings of the six endpoints, and a fixed matching can be concurrent at only one point; hence $c(A) \le 15$. The last display follows on setting $|\mathcal{U}(A)| = q + 1$. □

Theorem 7.2 (Uniqueness of $q = 11$)**.** *Let q be a prime power. If a six-arc $A \subset \mathrm{PG}(2, q)$ satisfies $\mathcal{U}(A) = \mathcal{C}(\mathbb{F}_q)$ for some nonsingular conic $\mathcal{C}$, then $q = 11$. At $q = 11$ every such A is projectively equivalent to the Clebsch hexagon.*

Proof. Lemma 7.1 gives $c(A) = (q - 6)(q - 9)$ with $0 \le c(A) \le 15$. The prime-power condition therefore leaves only $q \in \{4, 5, 9, 11\}$: the values at 7 and 8 are negative, while every integer $q \ge 12$ gives $c(A) \ge 18$.

At $q = 4$, every six-arc is a hyperoval. Each point off it lies on three secants (count the six hyperoval points on the five lines through the point), so $\mathcal{U}(A)$ is empty. At $q = 5$, every six-arc is an oval and hence a nonsingular conic; the standard internal/external point counts show that every off-conic point lies on a secant, so again $\mathcal{U}(A)$ is empty.

It remains to exclude $q = 9$. The argument has three steps: the six arc points must be internal to $\mathcal{C}$; passant joins of internal points become distance-two pairs in the Sylvester graph; and that distance graph has clique number five. Since every point of $\mathcal{C}$ is uncovered, no chord of A meets $\mathcal{C}$; every chord is therefore passant. A point off $\mathcal{C}$ lies on five passants if it is internal and four if it is external; hence the five chords through each vertex force all six vertices of A to be internal. Let Σ be the graph on the 36 internal points, with $P \sim Q$ when $Q \in P^{\perp}$ under the conic polarity. Standard conic counts give the intersection array

$$\{5, 4, 2; 1, 1, 4\},$$

so Σ is the Sylvester graph [13, §13.1.2]; see also the combinatorial uniqueness proof in [14, Thm. 6]. For distinct internal points, the unique possible common neighbor is $P^{\perp} \cap Q^{\perp} = (PQ)^{\perp}$; consequently PQ is passant exactly when P, Q are at distance two in Σ. Thus A would be a six-clique in the graph joining pairs at distance exactly two in Σ. But its clique number, denoted $\mathrm{eq}_2(\Sigma)$ in [12, Table 1], is five, a contradiction.

Therefore $q = 11$. The final assertion follows from Theorem 5.3, which puts every matching $q = 11$ representative in the single Clebsch orbit. $\square$

Proposition 7.3 (The Clebsch-family uncovered formula)**.** *Let K be a Clebsch hexagon over $\mathbb{F}_q$. Then*

$$|\mathcal{U}(K)| = q^2 - 14q + 45.$$

If $q \equiv 3 \pmod 4$ and $\mathcal{C}$ is Dye's associated conic, then $\mathcal{C}(\mathbb{F}_q) \subseteq \mathcal{U}(K)$ and

$$|\mathcal{U}(K) \setminus \mathcal{C}(\mathbb{F}_q)| = (q - 4)(q - 11).$$

In particular, among Clebsch hexagons in this congruence class, the associated conic is the complete projective deep-hole syndrome locus if and only if $q = 11$.

Proof. Dye defines a Clebsch hexagon by the occurrence of exactly ten Brianchon points, and his Theorem 1 classifies their existence and projective orbit [15, p. 271; Theorem 1, pp. 275–276]. These points are precisely the off-vertex triple-chord concurrences counted by $c(K)$. Thus $c(K) = 10$, and Lemma 7.1 gives

$$|\mathcal{U}(K)| = q^2 - 14q + 55 - 10 = q^2 - 14q + 45.$$

Dye also shows that the edges of K are non-secants of the associated conic when $q \equiv 3 \pmod 4$ [15, discussion preceding Theorem 6, pp. 281–282]. Hence every rational point of $\mathcal{C}$ is uncovered. Subtracting $|\mathcal{C}(\mathbb{F}_q)| = q + 1$ from the first formula gives

$$q^2 - 15q + 44 = (q - 4)(q - 11).$$

The factor vanishes only at $q = 4$ and $q = 11$, and the stated congruence leaves only $q = 11$. Conversely, Clebsch hexagons exist over $\mathbb{F}_{11}$ by Dye's Theorem 1, and the inclusion above is then an equality because both sets have twelve points. $\square$

Remark 7.4 (The next admissible field)*.* At $q = 19$, the same icosahedral construction still produces a Clebsch six-arc disjoint from its associated conic, but the covering fails. The twenty conic points remain uncovered [15, p. 281], while Proposition 7.3 gives

$$|\mathcal{U}(A)| = 140 = 20 + 120.$$

Thus $120 = (19 - 4)(19 - 11)$ additional points escape all fifteen secants, and $\mathcal{U}(A)$ lies on no conic. The configuration persists; exact conic filling is what the factorization isolates at $q = 11$.

7.1 The two reflection-arrangement exceptions

The chord moments below classify the neighboring arc sizes. The two survivors have a common arrangement-theoretic organization, although the arrangement calculation alone gives complement sizes, not conic equations.

Let F be a field of characteristic different from two containing τ with $\tau^2 = \tau + 1$. In dual coordinates, take the fifteen line normals

$$(1,0,0),(0,1,0),(0,0,1), \qquad \mathrm{cyc}(1, \epsilon\tau, \delta(\tau-1)) \quad (\epsilon, \delta \in \{\pm 1\}).$$

Over the real embedding with $\tau = (1+\sqrt{5})/2$, their signed versions are the roots of type H_3; over other fields this denotes the displayed finite reduction of the icosahedral mirror arrangement.

Proposition 7.5 (The Clebsch secants as reduced H_3 mirrors)**.** *The six fivefold points are*

$$(0,1,1-\tau),(0,1,\tau-1),(1,1-\tau,0),(1,\tau-1,0),(1,0,-\tau),(1,0,\tau).$$

They form an arc, their fifteen joins are the mirrors, and the other singularities are ten triple and fifteen double points. Over $\mathbb{F}_{11}$, take $\tau = 8$. The matrix

$$T = \begin{pmatrix} 2 & 3 & 8 \\ 10 & 6 & 9 \\ 2 & 2 & 5 \end{pmatrix}$$

maps these six column vectors onto the displayed Clebsch hexagon A. Writing line coefficients as rows, the induced map $\ell \mapsto \ell T^{-1}$ carries the mirrors onto the secants of A.

Proof. Pairwise cross-products of the normals give the stated $6_5, 10_3, 15_2$ ledger, and joining the six fivefold points recovers all fifteen normals. These are symbolic identities in $R = \mathbb{Z}[\tau]/(\tau^2-\tau-1)$, obtained by reducing each coordinate to $a + b\tau$. In the same ring, every three-by-three determinant on the six points is $2u$ for some $u \in R^\times$. Under any specialization $R \to F$, the image of u remains a unit and the image of 2 is nonzero when $\operatorname{char} F \neq 2$; hence the six points remain an arc. Finally, $\det T = 3$ in $\mathbb{F}_{11}$ and direct multiplication gives the two asserted set equalities. $\square$

Characteristic two is bad because the signs coalesce. In every other characteristic containing τ, the displayed lattice survives. In characteristic five, the reflection group $W(H_3)$ has order 120, which is divisible by the characteristic, so the modular group algebra is not semisimple. This does not collapse the coordinate arrangement: at $\tau = 3 \in \mathbb{F}_5$, the fifteen distinct mirrors and the full $6_5, 10_3, 15_2$ ledger remain. The finite checks for this statement and the $\mathbb{F}_{11}$ projectivity are recorded in Section 13. For the central rank-three hyperplane arrangement, a point where m projective mirrors meet has Möbius value $m-1$. Thus

$$\chi_{H_3}(t) = t^3 - 15t^2 + 59t - 45 = (t-1)(t-5)(t-9).$$

Dividing the affine complement count by $q-1$ gives $(q-5)(q-9)$ projective points. At $q = 11$, the points lying on respectively zero, one, two, and three mirrors number $12, 90, 15, 10$; here "one" means an ordinary mirror point. With the six fivefold points marked as columns, these are exactly the projective strata behind Proposition 4.5; the general passage to coset-leader multiplicities is due to Jurrius and Pellikaan [19].

For the standard four-frame $E = \{e_1, e_2, e_3, (1,1,1)\}$, the six joins have equations

$$X, Y, Z, X-Y, X-Z, Y-Z = 0.$$

These are the essentialized braid mirrors $x_i = x_j$ of type A_3, after setting $x_4 = 0$. Their four triple and three double points give

$$\chi_{A_3}(t) = (t-1)(t-2)(t-3), \qquad |\mathrm{PG}(2,q) \setminus \bigcup A_3| = (q-2)(q-3).$$

In the faithful-reduction range just stated for the explicit H_3 model, the two size equations are therefore

arc	secant arrangement	$\lvert\mathcal{U}\rvert = q+1$
four-frame	A_3	$(q-1)(q-5) = 0$
Clebsch hexagon	H_3	$(q-4)(q-11) = 0.$

The extraneous root $q = 4$ lies precisely in the bad characteristic-two regime, where the displayed H_3 reduction degenerates. Thus, in the faithful-reduction range, the arrangement factorizations leave precisely $q = 5$ and $q = 11$.

Theorem 7.6 (Conic-filling loci for $4 \le k \le 7$)**.** *Let A be a k-arc in $\mathrm{PG}(2,q)$, where q is a prime power and $4 \le k \le 7$. If $\mathcal{U}(A)$ is the full set of $\mathbb{F}_q$-rational points of a nonsingular conic, then*

$$(k,q) = (4,5) \qquad \textit{or} \qquad (k,q) = (6,11).$$

In the first case A is a projective four-frame; in the second it is a Clebsch hexagon. Conversely, both cases occur.

Proof with exhaustive finite verification. Let n_i count off-arc points incident with exactly i chords. For $k \le 7$, at most three pairwise disjoint chords concur. Counting chord–point incidences and pairs of disjoint chords gives the two universal moments

$$\sum_i i n_i = \binom{k}{2}(q-1), \qquad \sum_i \binom{i}{2} n_i = 3\binom{k}{4}.$$

For $k = 4$, every arc is a projective frame, so the A_3 calculation above gives $|\mathcal{U}(A)| = (q-2)(q-3)$. Equating this with $q+1$ gives $q = 5$; for the standard frame a direct substitution further gives

$$\mathcal{U}(A) = V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

where $V(f)$ denotes the projective zero set of f; this is a nonsingular six-point conic. For $k = 5$, the same moments force $n_2 = 15$ and $|\mathcal{U}(A)| = q^2 - 9q + 21$; equality with $q+1$ would require $q^2 - 10q + 20 = 0$, which has no integral root. The case $k = 6$ is Theorem 7.2 together with Theorem 5.3.

For $k = 7$, the moments give

$$|\mathcal{U}(A)| = q^2 - 20q + 120 - n_3.$$

If this equals $q+1$, then

$$n_3 = q^2 - 21q + 119, \quad n_2 = 105 - 3n_3, \quad n_1 = 3(q^2 - 14q + 42).$$

The standard arc-size bound gives $q \ge 7$. The condition $n_2 \ge 0$ gives $q \le 15$, so the prime-power candidates are $q \in \{7, 8, 9, 11, 13\}$; then $n_1 \ge 0$ eliminates $7, 8, 9$. Thus only $q = 11$ and $q = 13$ remain, with forced spectra $(n_1, n_2, n_3) = (27, 78, 9)$ and $(87, 60, 15)$ respectively. An exhaustive four-frame-normalized enumeration now supplies the finite exclusion. It finds 10232 distinct normalized seven-arcs at $q = 11$, of which 140 have $|\mathcal{U}(A)| = q+1$, and 53960 at $q = 13$, of which 1680 have that size. Every seven-arc is reached because it contains a four-frame; extending each normalized six-arc by its uncovered points produces every resulting seven-arc exactly three times, according to which non-frame point is deleted. For all 1820 candidates of the required size, the quadratic evaluation matrix has full rank six, so none lies on a conic. □

For $k \geq 8$ the second moment leaves additional concurrency parameters, so the analogous classification remains open.

8 Factorization memory in the conic ideal

Write the standard conic as $Q = XZ - Y^2$ and parametrize it by $\nu(s:t) = (s^2 : st : t^2)$. For $a = (a_0 : a_1)$ and $b = (b_0 : b_1)$ on $\mathrm{PG}(1, q)$, let L_{ab} denote the secant through $\nu(a), \nu(b)$, scaled so that

$$L_{ab}(\nu(s:t)) = [a, (s:t)]\,[b, (s:t)], \qquad [a, b] = a_0 b_1 - a_1 b_0.$$

For four endpoints, the Plücker relation gives the polynomial identity

$$L_{ab}L_{cd} - L_{ac}L_{bd} = [a, d][b, c]\,Q. \tag{9.1}$$

Thus a switch of two matching edges changes the secant product by a multiple of Q. Iterating switches proves equality of the restrictions for matchings in the generated component. The formal statement used here proves complete connectivity for the four-endpoint base and reversibility for a switch of arbitrary size; the finite Coxeter matching orbits are checked separately. We do not infer unrestricted connectivity of every perfect-matching graph from that base case.

Fix a reference matching M_0 and define

$$\Phi(M) = \frac{P_M - P_{M_0}}{Q}.$$

Changing M_0 translates every point by the same vector, so affine moment statements and difference ranks are independent of that choice. Kernel reduction of the three displayed Coxeter matching orbits gives

$$\begin{array}{r|ccc} & A_3 & B_3 & H_3 \\ \hline \text{matching-orbit size} & 5 & 14 & 22 \\ \operatorname{rank}\langle \Phi(M) - \Phi(N) \rangle & 3 & 6 & 10. \end{array} \tag{9.2}$$

The symbolic conic Laplacian decomposes the degree-$1, 2, 4$ homogeneous pieces into harmonic and radial parts at the stated characteristics; (9.1) is the bridge placing switch differences in the radial summand. The symbolic dimension formulas do not imply the finite factorization-image ranks in (9.2); those ranks require the displayed coordinate calculation.

Proof of Theorem 2.2(B1–B2). The pullback formula and Plücker identity prove (B1). The generic augmentation-kernel calculation controls the one-dimensional translation lost by choosing M_0. Exact row reduction of the three coordinate configurations gives (9.2), proving (B2). The trust table in Section 13 separates the symbolic identity, the kernel-checked finite ranks, and the classical identification of the coordinate orbits with the named Coxeter matchings.

9 Balanced sheets and cubic-first orientation

Let $\Omega = \Omega_+ \sqcup \Omega_-$ be the B_3 or H_3 factorization orbit, with $|\Omega_+| = |\Omega_-| = q$, and let $x_\omega = \Phi(\omega)$. For a signed weight $\epsilon = +1$ on Ω_+ and -1 on Ω_-, define a scalar shadow of the jth signed tensor moment by

$$M_j(\ell) = \sum_{\omega\in\Omega} \epsilon(\omega)\, \ell(x_\omega)^j.$$

The tensor moment is zero exactly when every multilinear shadow is zero. An affine change $x \mapsto Ax + c$ changes the first surviving degree-j moment by $A^{\otimes j}$; lower vanishing removes the translation.

The degree-two recovery is most transparent in the function space on the two sheets. Let L be the affine-evaluation subspace and $L^{\circ 2}$ the span of coordinatewise products of pairs of functions in L. The finite decoders on each sheet prove

$$L^{\circ 2} = \left\{ (u,v) : \sum_{\Omega_+} u = \sum_{\Omega_-} v \right\}. \tag{10.1}$$

The annihilator of the hyperplane on the right is the line spanned by the sheet-sign vector. Therefore a pointwise $\{\pm 1\}$ weight annihilates all quadratic products precisely when it is the sheet sign or its negative. This proves that the two displayed sheets are the unique balanced complementary halves, up to interchange.

An involution exchanges Ω_+ and Ω_- and negates the six compressed profile vectors. Consequently every even signed profile moment vanishes. On the positive profiles the primitive relation is

$$v_1 + 4v_4 + 6v_6 = 0, \tag{10.2}$$

which kills the signed barycentre. A displayed cubic coordinate is nonzero in both the B_3 and H_3 configurations. Hence degrees one and two vanish while degree three survives. This is the exact meaning of “cubic-first”: it concerns signed tensor moments and their linear functionals, not all conceivable invariants.

The signed cubic is fixed by the subgroup preserving each sheet and negated by the outer generator. The finite affine-action checks, together with nonvanishing, identify its stabilizer inside the displayed generated action with the kernel of the sign character. Identifying that generated action with the classical $\mathrm{PSL}_2(q)$ subgroup of $\mathrm{PGL}_2(q)$ is a named group-theoretic input.

Proof of Theorem 2.2(B3) and Theorem 2.3(C1–C2). Equation (10.1) and its annihilator calculation prove balanced uniqueness. The antipodal identity and (10.2) prove the even and linear cancellations. For the displayed profile coordinate, the first possible odd survivor is

$$2\big((-6)^3 + 4(-3)^3 + 6(3)^3\big) = 6 \neq 0 \pmod{11},$$

so degree three survives. The relative-invariant lemma gives the kernel stabilizer. All statements remain valid after an affine change of the quotient coordinates by the first-survival formula.

10 Six depth profiles and parent reconstruction

Put $G = \mathrm{PGL}_2(11)$, let $H = A_5$ stabilize one golden matching, and let $K = A_4$ be the common subgroup selected by an edge decoration. The finite two-sided action has six K–H

relation cells. Its K-orbits on each H-sheet have sizes $1, 4, 6$. One secant-incidence recount on a representative of each cell gives the six distinct profiles

$$v_1,\ v_4,\ v_6,\ -v_1,\ -v_4,\ -v_6. \tag{11.1}$$

The positive three satisfy (10.2) and two plane equations. The resulting linear map from the six relation indicators has rank two and kernel dimension four, but (11.1) shows that its restriction to the set of six labels is injective. Linear rank loss therefore does not imply loss of the finite label.

The singleton cells supply the reconstruction. The positive and negative singletons identify the two golden matchings, so without a choice the profile data recover their unordered pair. Choosing a sheet and its singleton selects the matching decoration required by the explicit parent reconstruction map. The conclusion is a parent with that decoration; an unmarked conic alone supplies neither the sheet nor the singleton.

The characteristic-eleven permutation module gives an intrinsic explanation of the rank drop. On one sheet the degree-eleven permutation module is the projective cover $P(1)$ with Loewy dimensions $1 \mid 9 \mid 1$. Its A_4-fixed space has orbit-sum basis of sizes $1, 4, 6$, and quotienting by the all-ones socle gives precisely the depth plane. The unique primitive positive dependence of the three rays is $1 : 4 : 6$, so the rays themselves recover both those orbit sizes and, by orbit–stabilizer, the orders $12, 3, 2$.

Proposition 10.1 (Modular depth quotient and source boundary)**.** *Over $\mathbb{F}_{11}$ the depth plane is*

$$P(1)^{A_4}/\operatorname{soc} P(1), \qquad \operatorname{Loewy}(P(1)) = 1 \mid 9 \mid 1.$$

Separately, the three-dimensional relative-cubic invariant space R and its coinvariants M_G have canonical maps

$$R \xrightarrow{\pi} M_G \xrightarrow{N} R$$

of ranks two and one, with $\operatorname{im}\pi = \ker N$ and $\operatorname{im} N = \ker \pi = \langle [1:3:9] \rangle$. Invariant-to-coinvariant projection and contraction by the unique rank-one semi-invariant form therefore construct the same canonical Tate two-plane. This plane is not naturally identified with the depth plane: the labelled cubic orbit classes have relation $[2, 9, 1]$, while the depth profiles have relation $[2, 8, 1]$; divided transfer kills the former balanced relations and fixes the latter socle.

Proof of Theorem 2.3(C3–C6) and Proposition 10.1. The exact generator permutations partition all twenty-two matchings into the six orbits of (11.1); the representative recounts are constant on those orbits. Row reduction gives the rank and kernel dimensions, and direct comparison proves set-theoretic separation. The two singleton recovery maps give the unordered matching pair, after which the matching-decorated recovery theorem returns the parent. Projectivity, the local two-dimensional commuting algebra, and exactness of A_4-fixed points give the $1 \mid 9 \mid 1$ quotient: the A_5 stabilizer has order prime to eleven, so its induced degree-eleven permutation module is projective; two-transitivity makes its commuting algebra $\mathbb{F}_{11}[I, J]$, and $J^2 = 11J = 0$ makes that algebra local. The module is therefore the indecomposable projective cover $P(1)$. Its three A_4-orbit sums have one-dimensional all-ones socle, and quotienting by it gives the rank-two depth plane. Since (10.2) spans the rational relation space and $\gcd(1, 4, 6) = 1$, it is the unique primitive positive dependence.

For the source boundary, the exact matrices give $N\pi = 0$ with ranks two and one on three-spaces. Hence $\operatorname{im}\pi \subseteq \ker N$ and both sides have dimension two, so they are equal; the displayed one-dimensional image of N is the kernel of π. Finally, in the orbit-sum basis put $s = (1,1,1)^t$ and $B = s(1,4,6)$. Then $B^2 = 11B$; B kills every balanced relation and satisfies $Bs = 11s$. Divided transfer therefore kills the source relation but fixes the depth socle, proving the stated non-identification.

11 Rank-three arithmetic gluing

Lemma 11.1 (The split–inert frame calculation)**.** *The frame polynomials at the three working primes have the following factorizations:*

$$\begin{array}{r|c|c} A_3,\ q=5 & x^2-2 & \textit{irreducible}, \\ B_3,\ q=7 & x^2-2=(x-3)(x-4) & \textit{split}, \\ H_3,\ q=11 & T^2-T-1=(T-4)(T-8) & \textit{split}. \end{array}$$

In the first row the two roots form one Frobenius orbit over $\mathbb{F}_{25}$; in the last two rows they are two rational points exchanged by the quadratic conjugation.

Proof. The nonzero squares modulo five are 1 and 4, so 2 is not a square. If $u^2 = 2$ in $\mathbb{F}_{25}$, the nontrivial automorphism of $\mathbb{F}_{25}/\mathbb{F}_5$ sends u to $-u$; it is Frobenius, hence the two roots form one orbit. Direct multiplication gives the two displayed split factorizations modulo seven and eleven. In each split quadratic the nontrivial algebra involution exchanges its two roots. □

The factorization split has a uniform arithmetic description at the three conic fields. The A_3 marker polynomial is inert over $\mathbb{F}_5$, and its two geometric roots form one Frobenius orbit; projectively the matching datum is fused. The B_3 and H_3 polynomials split over $\mathbb{F}_7$ and $\mathbb{F}_{11}$, producing the two silver and two golden sheets. Exact projective permutation closure verifies these three cases and the fixed-point-free matching involutions.

In the golden case, the two displayed stabilizers have order 60, their intersection has order 12, and the subgroup they generate has order 660. The rational rotation denoted R_z reduces to a projectivity of nonsquare determinant that exchanges the two matchings. Its common octahedral carrier has determinant kernel of order 12. With the classical identifications of these literal groups, this is the $A_5, A_4, \mathrm{PSL}_2(11), S_4$ gluing stated in Theorem 2.4(D2). The exact finite closure does not by itself prove the spinor-norm or number-field names attached to the rational model.

Above the fixed full-conic child, the twenty-two explicitly reconstructed parents form two eleven-element orbits under the determinant character. The quotient by the kernel action is a free two-point set. The same sign character controls the matching sheets, the outer transporter, and the fixed-child quotient. There is no invariant point, while a chosen point selects one sheet and hence one decorated parent.

Proof of Theorem 2.4. The frozen reduction polynomials, matching permutations, and group closures prove (D1) and the literal finite content of (D2). The determinant-kernel comparison proves equality of the three sign readouts. The fixed-child fibre computation gives the two eleven-element orbits and the free quotient, proving (D3). The classical group and rational spinor interpretations are precisely the cited-input boundary recorded in Section 13.

12 The survival and erasure ledger

The orientation torsor is not preserved by every natural passage. The following table states the exact Paper-1 boundary.

passage	retained data	verified scope
unmarked conic child	none	no preferred matching, parent, or orientation
quotient moments $1, 2$	unordered sheets	unique balanced complementary halves in the displayed B_3, H_3 configurations
signed cubic	orientation	first survivor among signed tensor moments and their linear shadows
singleton depth profile	decorated parent	the six displayed $A_4\backslash G/H$ labels
quadratic-residue/Witt design	finite shadow	exact incidence, code, full-support, secant, and Hadamard tables
golden reduction	conditional split/fusion	the exact frozen residue classes modulo 40
finite theta signatures	erased	equal sheet signatures in the displayed genus-three and genus-five tables
fixed-party quantum transport	erased	complete support transport and transition-forced local-Clifford search for the displayed states
ambient Fourier matrices	finite shadow	exact square, trace, and weighted-adjoint identities; no small-Weil-module identification

The design row includes the cyclic quadratic-residue two-design, the displayed ternary weight distribution, the Steiner 5-design on minimum supports, the twelve full-support projective points and all sixty-six of their secants, and the order-twelve Hadamard identity. Parity extension gives the Steiner system $S(5, 6, 12)$; the two degree-twelve M_{11} parents meet in the frozen $\mathrm{PSL}_2(11)$ and generate M_{12}, while Hadamard row–column duality exchanges them. These classical names are attached only after the literal finite closures and normalizer checks.

The modulo-40 statement is conditional on the frozen arithmetic predicates: within that domain the two golden sheets are visible or fused according to the exact residue partition checked by the certificate. It is not an all-prime reciprocity theorem, a density statement, or a construction of a common quadratic-character carrier.

Proposition 12.1 (The conditional mod-40 law)**.** *Let $\ell \neq 2, 5$ be an odd prime satisfying the frozen golden-marker hypotheses. Golden splitting is equivalent to $(5/\ell) = +1$. Among the split classes, the sheets are fused when $(2/\ell) = +1$ and visible when $(2/\ell) = -1$. Consequently the split-visible residue classes modulo 40 are*

$$11,\ 19,\ 21,\ 29,$$

and the split-fused classes are

$$1,\ 9,\ 31,\ 39.$$

The other eight unit classes modulo 40 are golden-inert.

Proof. Quadratic reciprocity gives $(5/\ell) = (\ell/5)$, so golden splitting means $\ell \equiv \pm 1 \pmod 5$. The supplementary law gives $(2/\ell) = +1$ exactly for $\ell \equiv \pm 1 \pmod 8$ and -1 exactly for $\ell \equiv \pm 3 \pmod 8$. Combining the two congruences by the Chinese remainder theorem gives the two displayed four-element lists. The remaining eight units modulo 40 have $(5/\ell) = -1$. □

The final two rows are likewise bounded. Their certificates check finite quadratic-value histograms and matching-intersection signatures; the divisor-theoretic theta interpretation uses the cited classical input. The quantum certificate proves the displayed monomial transport, signed-transpose identity, and complete fixed-party kernel–image equivalence. It does not classify arbitrary complex local unitaries or assert a canonical geometric reconstruction of every state.

The fixed four-sheet contraction has a sharper partial-retention boundary. It is not a step in the Clebsch-parent reconstruction above. Consider the ordered six-column pencil

$$\begin{aligned} H_t = \big((0,1,1-t),(0,1,t-1),(1,1-t,0),\\ (1,t-1,0),(1,0,-t),(1,0,t)\big). \end{aligned}$$

Write h_i for its columns and $g_i(x) = h_i^{\mathsf{T}} x$ for the corresponding coordinate covectors on the message space. The admitted locus removes the degenerate and GRS parameters by the nonvanishing condition in the theorem below. On that locus, the theorem explains exactly what the contraction retains: two resonance fibres and their signed refinement. Since it does not recover the pencil parameter, its role in Paper I is to mark a strict survival boundary rather than to add another reconstruction mechanism.

Theorem 12.2 (Four-sheet cover holonomy)**.** *Let V be the three-dimensional message space of the admitted non-GRS pencil in odd characteristic, with coordinate covectors $g_0, \dots, g_5$. Put*

$$A = -4t(t-1)^2, \quad B = (t^2-t+1)(t^2-3t+1), \quad w = B/A, \quad z = w^2,$$

and assume $2t(t-1)BG \neq 0$, where $G = t^4 - 4t^3 + 7t^2 - 4t + 1$. Join four left to four right copies by the six matchings $\sigma_0, \dots, \sigma_5$

$$1,\ (03)(12),\ (23),\ (02)(13),\ (01)(23),\ (021).$$

For each party assignment $p \in S_6$, put V on every vertex, $\mathbb{F}_q$ on every edge of color i, and use $g_{p(i)}$ as both restriction maps. Then:

(i) *constant sections form a three-space. After quotienting them, the 24-equation section problem is the kernel of a 9-by-9 operator on U^3, where $U = \mathbb{F}_q^4/\mathbb{F}_q(1,1,1,1)$; its nine blocks are two-matching holonomies;*

(ii) *a nonconstant generic section occurs precisely in one axial relative-frame type, of size 96, at $z = 2$, or in either of two signed six-cycle types, each of size 192, at $w = \pm 2/3$. Every generic excess kernel is one-dimensional, so the reduced rank-drop divisor is*

$$(z-2)(9z-4) = 0.$$

The common derived A_4 acts freely, giving 8 and 16 quotient classes;

(iii) *characteristic 7 merges the axial divisor with one signed divisor and gives $96 + 192 = 288$ active assignments. Characteristics $11, 13, 41$ only ramify the pullback to the t-line. Characteristic 3 removes both reduced components at the excluded boundary, while in characteristic 5 the signed component moves to that boundary and the axial component remains.*

The theorem detects these two reduced values; it does not recover the whole pencil or assert four-copy minimality.

Proof. A section is a tuple $(x_a, y_b) \in V^8$ satisfying

$$g_{p(i)}(x_a - y_{\sigma_i(a)}) = 0.$$

The diagonal tuples are the constant three-space. Subtract y_0 from all vertex values and then use code coordinates $3, 4, 5$ as free coordinates. If ρ_i is the action of the i-th matching on U, the remaining equations are

$$\sum_{k=0}^{2} Q_{ik}(\rho_i - \rho_{3+k})Y_k = 0, \qquad Q = \begin{pmatrix} a & b & c \\ a & c & b \\ d & d & d \end{pmatrix},$$

where

$$a = -2(t-1)^2, \quad b = -(t^2 - t + 1), \quad c = -(t^2 - 3t + 1), \quad d = -2(t-1).$$

Thus each block is the transport around the even cycles in the union of two matchings. Reversing the elimination recovers the original 24-by-21 quotient matrix, with the same excess kernel dimension.

The identities

$$a = b + c, \qquad d^2 = -2a, \qquad A = a(c-b), \qquad B = bc$$

reduce the signed cycle-cover ledger of this operator to

$$\begin{aligned} P_- &= AB(3B - 2A), \\ P_+ &= AB(3B + 2A), \\ P_2 &= A(B^2 - 2A^2). \end{aligned}$$

The remaining determinant factors are units on the admitted pencil. Hence the three localized obstructions are one-dimensional and give $w = 2/3, -2/3$ and $z = 2$. This derives the constants from the transport cycles rather than fitting them to the 720 ranks.

It remains to count the supporting assignments. The code columns and the reduced transport each carry a frame of three opposite pairs. The axial type is the relative-frame double coset with one marked common axis; its seam has order 8, so its size is $48 \cdot 16/8 = 96$. The two orientations of disjoint frames have seams of order 6, and each has size $24 \cdot 48/6 = 192$. The order-24 action here is the rotational octahedral action on six vertices, with point stabilizer C_4, and the axial C_2^3 seam has orbits $2 + 4$. These octahedral symmetries belong to the reduced linear transport frame, not to the bare bipartite multigraph. The common derived A_4 acts freely, proving the two quotient counts.

Finally put $r = b/c = (t + t^{-1} - 1)/(t + t^{-1} - 3)$. The three nonboundary root orbits are

$$r \in \{1/2, -2\}, \qquad r \in \{-1/2, 2\}, \qquad r^2 \in \{1/2, 2\}.$$

The sole nonboundary branch point is $t = -1$, where $r = 3/5$. Substitution gives the collision primes $7, 11, 13, 41$. In characteristic 7,

$$(3B - 2A)(3B + 2A) - 2(B^2 - 2A^2) = 7B^2,$$

so the axial and signed-union root schemes coincide after localizing at B, while their assignment supports remain disjoint. The other three primes double one pullback root without changing its reduced divisor or kernel dimension. Direct specialization puts both obstructions on the excluded boundary in characteristic 3, and only the signed obstruction there in characteristic 5. $\square$

Lemma 12.3 (Determinant-sign transport and the no-section obstruction)**.** *Let $\chi : G \to C_2$ be surjective and put $H = \ker \chi$. Then G/H is a free transitive C_2-torsor with no G-fixed point. More precisely, if X is any two-point G-set whose action factors through χ and whose induced C_2-action is nontrivial, then for each $x \in X$ there is a unique G-equivariant bijection*

$$G/H \longrightarrow X$$

sending H to x. Thus all such two-valued carriers are realizations of one torsor: comparing them requires exactly one marking bit, and no G-equivariant marking exists before that bit is adjoined.

Proof. The first isomorphism theorem identifies G/H with C_2 as a G-set, where G acts through the surjection χ. Surjectivity makes the nontrivial element exchange the two points, so neither is fixed. A G-equivariant choice from an unmarked singleton base would have fixed image and is impossible. Given $x \in X$, the only possible equivariant map is

$$gH \longmapsto g \cdot x.$$

It is well defined because H acts trivially, and it is bijective because the nontrivial coset exchanges the two points. Its value at H proves uniqueness. The two choices of x are therefore precisely the two possible markings. □

Theorem 12.4 (Torsor Rosetta: one class and one forced outer swap)**.** *Let T_q denote the determinant-sign torsor for the split rank-three working primes. The following statements hold.*

(E1) *For B_3 at $q = 7$ and H_3 at $q = 11$, T_q is a free C_2-torsor. For A_3 at $q = 5$, the marker polynomial is inert: the two spin lifts form one Frobenius orbit over $\mathbb{F}_{25}$, hence one connected étale point over $\mathbb{F}_5$ and no orientation bit.*

(E2) *The certified dictionary has twelve rows: the two matching sheets, the two nontrivial unipotent classes, the two cyclotomic period factors, the two primes of $\mathbb{Q}(\sqrt{-q})$, the two primes of $\mathbb{Z}[\sqrt{2}]$ or $\mathbb{Z}[\phi]$ above the working prime, the two lower Weil constituents, the two cubic signs, the two QR/Barker polarities, the two signed Fourier sectors, the outer M_{12} hinge, the characteristic-zero $\operatorname{Spec} \mathbb{Q}(\sqrt{5})$ row, and the Reed–Solomon fixed-child extremal fibre. Each two-valued row available at a given q is outer-equivariantly identified with T_q. The M_{12} row is the acting face of the same swap, not another two-point carrier; the characteristic-zero row specializes specifically to T_{11}.*

(E3) *The marked Coxeter matching points all available rows simultaneously through the single trace rule. After forgetting it there is no section: the Čech cocycle, the exact one-bit selector cost, and the free-torsor description are three readouts of*

$$[T_q] = \mathrm{sgn} \colon \mathrm{PGL}_2(q) \longrightarrow C_2, \qquad \ker(\mathrm{sgn}) = \mathrm{PSL}_2(q).$$

(E4) *At $q = 11$ the Mathieu hinge is forced outer. Indeed $N_{M_{12}}(\mathrm{PSL}_2(11)) = \mathrm{PSL}_2(11)$, $\mathrm{PGL}_2(11)$ is not a subgroup of M_{12}, and the two M_{11} parents are non-conjugate in M_{12}.*

(E5) *The cubic sign and trace residue co-vary with T_q, but their numerical dictionary is convention-dependent between $q = 7$ and $q = 11$. The uniform datum is the marking, not a residue formula. No integral Galois-odd cubic is assumed: the bit is carried arithmetically by the prime fibre.*

Proof. For every available row, the finite calculation verifies the same structural hypothesis: it is a two-point G-set and the action on it is the character sgn. Lemma 12.3 then forces its identification with T_q after one point is marked. This single quotient-action argument organizes all twelve rows. The matching, unipotent, period, split-prime, Weil, and reduction calculations verify that hypothesis for the first seven rows. Exact outer-element calculations verify it for the QR/Barker and signed Fourier rows. The normalizer and two-parent computations give the forced outer Mathieu clause. The characteristic-zero resolvent $\operatorname{Spec}\mathbb{Q}(\sqrt{5})$ reduces at eleven to the two golden fibres, and the fixed-child computation identifies its two-sheet quotient with the same matching orbit. The two comparison squares identify the Čech and selector certificates with the sign character. Finally, the explicit residues at seven and eleven prove the per-case-dictionary boundary. Every finite comparison and its independent replay is listed in Section 13. □

13 Verification architecture

Tables 2 and 3 separate mathematical input, kernel-checked formal proof, and exhaustive finite enumeration. Here "kernel-checked" means that Lean's trusted kernel checks the stated implication; executable replays instead enumerate the finite search space with exact arithmetic. The paper repository and the shared Lean repository are distributed separately. Paper paths below are relative to the paper root; Lean paths are relative to the shared Lean root. The claim-by-claim machine-readable ledger in `verification/trust_manifest.json` records the exact manuscript hash, shared-Lean commit, gate and audit hashes, executable artifacts, cited inputs, and residual trust for every published claim. From the paper root, the single release command is

```
python3 verification/verify_release.py -lean-root /path/to/shared-lean.
```

It validates the manifest before running any declared check, rejects a dirty or mismatched checkout, and verifies that the run changes neither repository.

Result	Formal gate	External semantic boundary or replay
pairing forgetting and harmonic quotient	`RelativeConicArcs/Gates/ClebschPaperTrust.lean`	standard conic identification; four-endpoint connectivity boundary
factorization ranks and balanced sheets	same paper trust gate	exact finite Coxeter coordinate identification
cubic and six-profile depth	same paper trust gate	classical $A_4, A_5, \mathrm{PGL}_2(11)$ names; finite relation-cell replay
arithmetic gluing and torsor dictionary	same paper trust gate	`verification/evidence/torsor_dictionary/verify.py`
Witt–Hadamard shadow and survival boundaries	same paper trust gate	literal finite actions; classical Mathieu and spinor names remain cited inputs
theta, Fourier, and fixed-party passages	same paper trust gate	`verification/evidence/passage_interfaces/verify.py`
four-sheet cover holonomy	conceptual sheaf and cycle proof in Theorem 12.2	`verification/evidence/four_sheet_holonomy/verify.py`

Table 1: Verification map for the factorization-memory results. Sharing a gate does not merge claim routes: the trust manifest names the adequate terminal and axiom list separately for each row.

Provenance and assistance disclosure. The author selected the mathematical claims, checked the cited sources, and accepts responsibility for the manuscript. Language models assisted with proof exploration, formal-code drafting, executable-checker drafting, and editorial revision. No model output is treated as evidence. Every paper-facing claim is routed in the trust manifest to a kernel-checked Lean statement, an exact executable replay, a mathematical proof with pinpointed classical inputs, or an explicit decomposition of those routes. Generated finite data are retained only when their generator, schema, coverage, hashes, replay, and residual trusted boundary are recorded.

Result	Mathematical input	Kernel-checked formalization
A_5 orbits and syndrome conic	Dye; Proposition 4.1	`RelativeConicArcs/Q11A5PointOrbits.lean`; `RelativeConicArcs/Q11Coding.lean`
A_3/H_3 arrangement synthesis	exact root coordinates; intersection lattices	`RelativeConicArcs/ReflectionArrangements.lean`; `RelativeConicArcs/ReflectionArrangementDecoding.lean`
Decoding and support bipartition	arc–coset dictionary; Edge–Dye triangles	`RelativeConicArcs/Q11DecodingSynthesis.lean`
Rigidity theorem	line bound; chord defect; Dye's Theorem 1	`RelativeConicArcs/Q11DyeAxioms.lean`; `RelativeConicArcs/Q11DyeConsequences.lean`; `RelativeConicArcs/SixArcDefectBridge.lean`
Gap and low-degree loci	finite linear algebra over $\mathbb{F}_{11}$	—
Uniqueness of $q = 11$	chord moments; cited Sylvester bound	`RelativeConicArcs/ClebschChordDefect.lean`; `RelativeConicArcs/Q9Sylvester.lean`
Clebsch-family formula	Dye's ten Brianchon points and edge criterion	—
The range $4 \le k \le 7$	two universal chord moments	`RelativeConicArcs/SmallKChordMoments.lean`; `RelativeConicArcs/SmallKGeometricBridge.lean`

Table 2: Mathematical and formal dependencies of the principal results. An em dash means that the manuscript makes no Lean-formalization claim for that row; its executable evidence, where applicable, is listed separately in Table 3.

Result	Exhaustive or independent executable check
Syndrome conic and rigidity census	`check_rigidity_degenerate_conic.py`; 1548-representative frame-normalized enumeration
A_3/H_3 arrangement synthesis	`check_reflection_arrangements.py`
Decoding and support bipartition	`check_decoding.py`; `check_chirality.py`; `check_code_automorphisms.py`
Gap and low-degree loci	`check_global_conic_gap.py`; `check_perturbation_gap.py`; `check_low_degree_loci.py`; `check_low_degree_loci.sh` (invokes `check_low_degree_loci.sing`)
Uniqueness of $q = 11$	`check_small_q_uniqueness.py`; independent $q = 9$ construction
Clebsch-family formula	`check_q19_nonexample.py`
The range $4 \le k \le 7$	`check_small_k_conic_filling.py`

Table 3: Executable checks, separate from the formal dependencies in Table 2 and organized by replay rather than row-for-row parallelism with that table.

For the reflection-arrangement row, the Python replay checks the finite $\mathbb{F}_5/\mathbb{F}_{11}$ specializations independently. The Lean file checks the $\mathbb{F}_{11}$ coordinate projectivity and pointwise secant-index transport, the characteristic-two boundary, both finite intersection spectra, and the displayed polynomial factorizations. These finite checks are separate from the symbolic R-calculation in Proposition 7.5.

The formal development axiomatizes exactly two consequences of Dye's Theorem 1: the ten-point Brianchon bound and the classification of equality. The line bound and chord-defect identity are proved independently. At $q = 11$, the finite census also checks the imported numerical conclusion: $|\mathcal{U}(A)| = 12 = 22 - 10$ is equivalent to the maximal value $c(A) = 10$.

13.1 Independent small-field census

Normalizing an ordered four-frame and enumerating the remaining unordered pair gives the following exact check.

q	representatives	histogram of $\lvert\mathcal{U}(A)\rvert$	conic matches
2	0	—	0
3	0	—	0
4	1	0^1	0
5	3	0^3	0
7	70	$0^{40}, 2^{30}$	0
8	195	$0^{45}, 4^{150}$	0
9	441	$0^6, 4^{15}, 6^{120}, 7^{120}, 8^{180}$	0
11	1548	$12^6, 16^{30}, 18^{150}, 19^{300}, 20^{630}, 21^{360}, 22^{72}$	6
13	4015	$36^{85}, 38^{210}, 39^{480}, 40^{1080}, 41^{1800}, 42^{360}$	0

Table 4: Small-field census. "Representatives" means frame-normalized point sets, not projective classes; m^r means that r representatives have $|\mathcal{U}(A)| = m$. The six conic matches at $q = 11$ form one projective class.

The enumeration uses exact arithmetic in every field and tests both nonsingular and singular conic forms in characteristic two. A separate $q = 9$ certificate constructs the polarity graph, verifies the passant/distance-two dictionary, and computes clique number five.

Appendix: The relative-cubic Tate plane

This appendix records the source-side naturality statement kept separate from the reconstruction proof. Put $G = \mathrm{PGL}_2(11)$ and $M = \mathrm{Sym}^3(W) \otimes \chi$, where χ is the outer sign. The invariant space $R = M^G$ and the coinvariant space M_G both have dimension three. Canonical projection and group norm form the cycle

$$R \xrightarrow{\pi} M_G \xrightarrow{N} R, \qquad \operatorname{rank} \pi = 2, \quad \operatorname{rank} N = 1,$$

with $\operatorname{im} \pi = \ker N$ and $\operatorname{im} N = \ker \pi = \langle [1 : 3 : 9] \rangle$. Hence $R/\ker \pi$ is a canonical Tate two-plane. Independently, the invariant symmetric-form pencil on W has a unique rank-one member, the square of an outer-odd covector. Contraction against that covector has matrix

$$\begin{pmatrix} 8 & 1 & 0 \\ 0 & 8 & 1 \end{pmatrix}$$

in the frozen relative basis, so it has the same kernel and constructs the same plane.

Indeed $N\pi = 0$ makes $\operatorname{im}\pi$ a subspace of $\ker N$; the two spaces have the same dimension two, proving equality. The first isomorphism theorem then identifies $R/\operatorname{im} N = R/\ker\pi$ with $\operatorname{im}\pi = \ker N$. The contraction matrix has rank two and kernel $[1 : 3 : 9]$, giving the second construction without fitting an isomorphism.

These two source constructions do not identify the Tate plane with the depth plane. The three canonical A_4 orbit-cube classes have sole relation $[2, 9, 1]$, whereas the labelled depth profiles have relation $[2, 8, 1]$. The integral divided-transfer operator kills the balanced source relations but fixes the depth socle. Orbital, invariant-correlation, quadratic-coinvariant, and ordered-rank-flag tests likewise leave no canonical bridge. Thus the positive statement is source intrinsicity; the non-identification is a proved scope boundary.

Appendix: Headline statement adequacy

The deterministic extraction in `verification/statement_adequacy.json` preserves the exact TeX of every theorem-like environment, its label or content-addressed key, source line, and SHA-256 digest. The following printed summary records the intended adequacy boundary of the headline blocks; the machine-readable trust manifest splits every numbered clause into its own route.

Statement	Formal/computational surface	Clauses kept external or mixed
Theorem 2.1	retained rigidity gates; weighted-adjoint and Coxeter-phase terminals	Dye equality inputs; coordinate-to-Coxeter and extended-GRS identifications
Theorem 2.2	conic-matching, harmonic, factorization, and balanced-sheet terminals	arbitrary-size matching connectivity beyond the proved boundary; classical group identification
Theorem 2.3	moment-trade, balanced-sheet, double-coset depth, and modular-depth terminals	named $A_4, A_5, \mathrm{PGL}_2(11)$ semantics; no identification of the depth plane with the relative-cubic plane
Theorem 2.4	arithmetic-gluing and fixed-child terminals plus exact replay	classical spinor and number-field interpretations
Theorem 12.4	torsor dictionary, survival-boundary, and Mathieu-hinge terminals plus independent exact replay	classical group and cohomological names; no cross-characteristic scheme, integral cubic, or all-prime classification

Table 5: Human-readable headline adequacy summary. The exact terminal, axiom, file-hash, citation, and residual-trust data are in the manifest.

14 Conclusion

The projective deep-hole conic of the Clebsch code is both rigid and forgetful. As an unmarked locus it characterizes the parent among six-arcs of $\mathrm{PG}(2, 11)$, yet as a conic code it retains no preferred secant pairing or orientation. The conic ideal resolves this tension. Secant-product differences vanish on the child but survive after division by its equation; their quotient configuration recovers the unique unordered sheets from second moments, orients them in degree three, compresses them to six double-coset profiles, and recovers a decorated parent from the singleton fibres.

The rank-three Coxeter phase explains why this chain belongs to a complete small family. The reflection-complement codes for A_3, B_3, H_3 obey one distance law and become conic codes at $q = h + 1$. Their matching data fuse in the A_3 control, split through the silver roots in B_3, and split through the golden roots in H_3. In the last case the two stabilizers meet along A_4, generate $\mathrm{PSL}_2(11)$, and are exchanged through the rational S_4/A_4 hinge.

The survival ledger gives a precise boundary. Quadratic-residue designs and perfect-code spans retain finite shadows, and the frozen golden reduction data obey the exact modulo-40 partition. The displayed theta signatures do not orient the sheets, and the displayed golden quantum states are equivalent even with fixed party labels. The Fourier matrices satisfy the exact restricted identities, but this does not turn the quotient spaces into small Weil modules. Orientation is carried by specific comparison data, not by every natural invariant of the objects. Likewise, the four-sheet contraction retains only the two resonance fibres of Theorem 12.2; it does not recover its pencil and does not participate in parent reconstruction.

The torsor-Rosetta theorem identifies both the object and its finite mechanism. At the split working primes the missing bit is obtained by determinant-sign descent: every available two-valued carrier is a G-set with the same quotient character, so the transport lemma forces all of them to be realizations of T_q. At the inert A_3 prime the bit fuses. The characteristic-zero golden resolvent reduces to T_{11}, the fixed-child fibre is the same finite torsor, and the Mathieu action is forced outer. The one no-section class has cohomological, selector, and torsor readouts. No integral Galois-odd cubic is needed.

The remaining question is therefore deliberately narrow: which arithmetic or metaplectic mechanism produces the characteristic-q gluing of this torsor while the theta and quantum passages erase it?

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